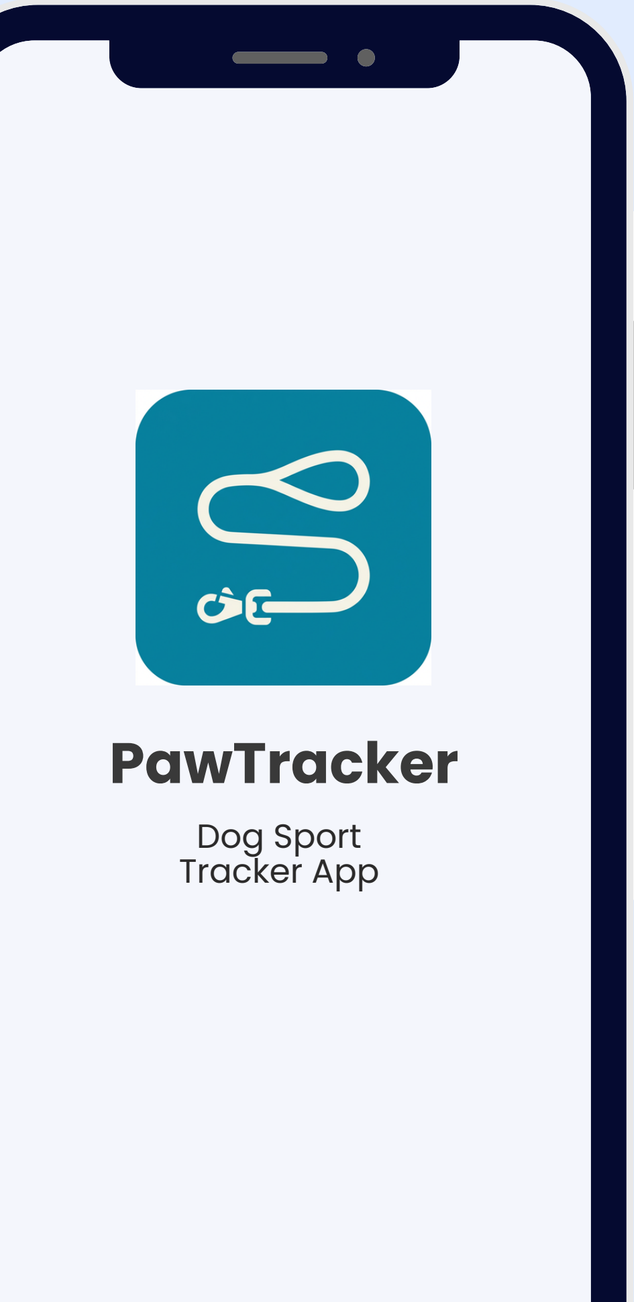

PawTracker

Dog Sport Tracker App

Mobile Application Development Project

Emmi Korhonen & Farnaz Jamilpanah

Introduction

Problem

Dog owners often don't know if their pets are getting enough physical activity → Might affect negatively to dog's health.

It's hard to track walking routes, distances, and durations manually.

Solution

PawTrack Pet Sport Tracker provides a simple, visual interface that shows dog's walking routes

Uses the phone's GPS to automatically record walking routes, calculate distance and duration and store the data for long-term tracking.

Target Users: Active dog owners who are careful or concerned about their pet's health. They are aomfortable using smartphones.

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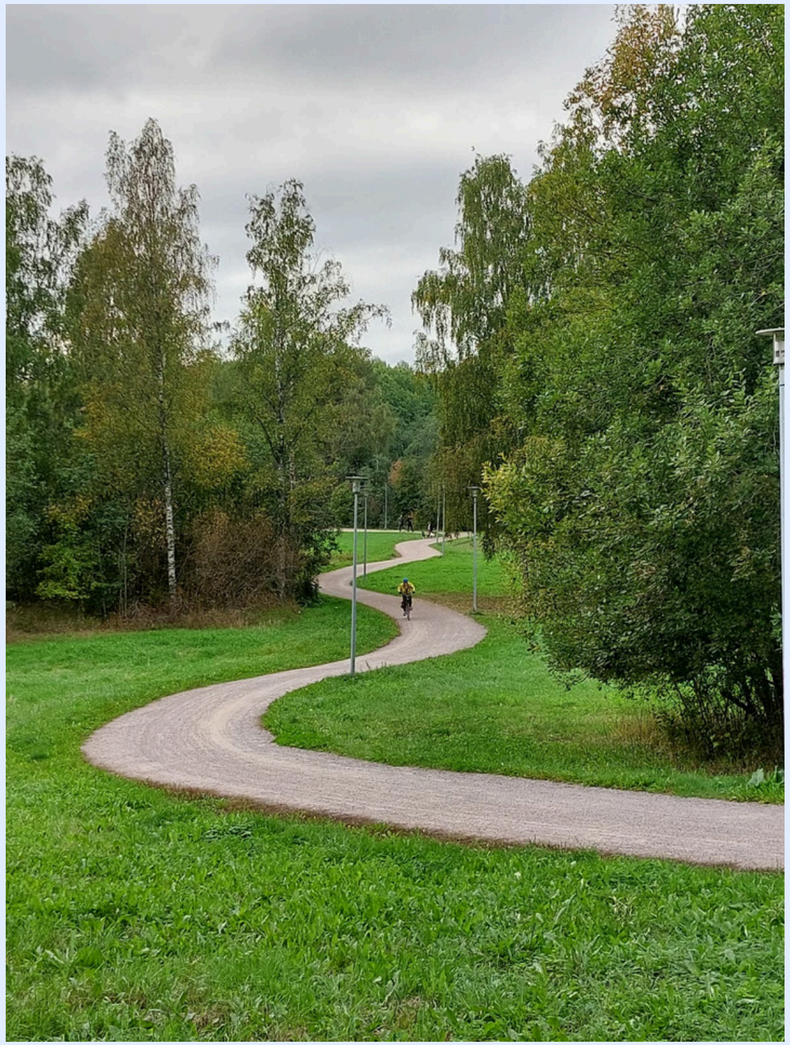
Core Features (MVP)

- **GPS tracking:** Records the pet's walking route in real time
- **Map view:** Shows the route visually on a map
- **Activity statistics:** Distance, duration and daily/weekly totals
- **History:** Stores past walks using Room database
- **Design UI:** Clean, accessible, and responsive interface
- **Unit tests:** Logic tests for ViewModel and data handling

Optional Features

- **Dog profile:** Stores dog information and profile picture
- **Dog breed API:** Information about different breeds and their recommended activity levels
- **Health log:** Allows owners to record observations about the dog's droppings for health monitoring
- **Weather API:** Shows weather
- **Notifications:** Reminds the user when the dog hasn't been active enough
- **AI:** Provides simple weekly activity analysis



Technology Stack

Technology Stack

- **Programming Language:** Kotlin
- **UI Framework:** Jetpack Compose
- **Architecture:** MVVM (ViewModel + StateFlow)
- **Database:** Room
- **Sensors:** GPS (Location Services)
- **APIs:** Google Maps, OpenWeather API (optional)
- **Testing:** JUnit (unit tests), UI tests

Development Tools

- Android Studio
- GitHub
- Jira
- Figma
- Adobe Firefly

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Application Architecture

Each screen has its own ViewModel and UiState

UI observes changes from the ViewModel and updates automatically

```
├ ui/
│   ├ main/
│   │   ├ MainScreen.kt
│   │   ├ MainViewModel.kt
│   │   └ MainUiState.kt
│   │
│   ├ tracking/
│   │   ├ TrackingScreen.kt
│   │   ├ TrackingViewModel.kt
│   │   └ TrackingUiState.kt
│   │
│   ├ history/
│   │   ├ HistoryScreen.kt
│   │   ├ HistoryViewModel.kt
│   │   └ HistoryUiState.kt
│   │
│   ├ components/
│   │   ├ Buttons.kt
│   │   ├ Cards.kt
│   │   ├ Charts.kt
│   │   ├ InputFields.kt
│   │   └ ListItems.kt
│   │
├ navigation/
│   ├ ScreenNav.kt
│   └ Screens.kt
```

```
├ data/
│   ├ local/
│   │   ├ DogEntity.kt
│   │   ├ WalkEntity.kt
│   │   ├ GPSEntity.kt
│   │   ├ DogDao.kt
│   │   ├ WalkDao.kt
│   │   └ AppDatabase.kt
│   │
│   ├ remote/
│   │   ├ WeatherApi.kt
│   │   ├ RetrofitInstance.kt
│   │   └ WeatherResponse.kt
│   │
│   └ repository/
│       ├ DogRepository.kt
│       ├ WalkRepository.kt
│       ├ GPSRepository.kt
│       └ WeatherRepository.kt
│
├ location/
│   └ GPSService.kt
│
├ utils/
│   ├ Extensions.kt
│   └ Constants.kt
```

Room stores walk history and statistics

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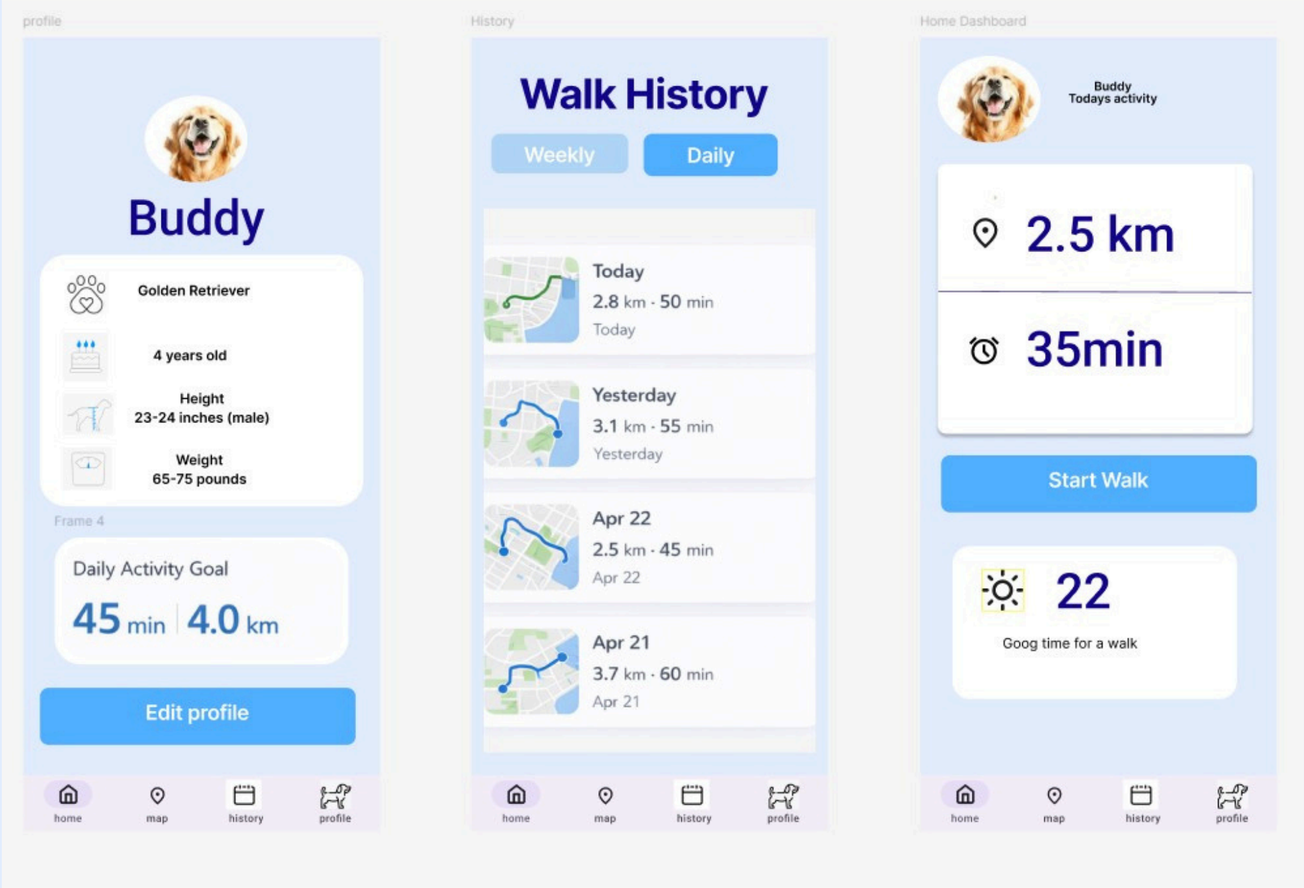
Figma Design



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Scrum Process

Approach

Project duration 18.3-6.5.2026

- Sprint length: 1 week
- Total: 6 sprints

Communication

- Daily Scrum: Mon, Wed, Fri
- Ongoing discussion on Discord

Tracking

- Product backlog
- Burndown chart

Roles

Product Owner:
Emmi and Farnaz?

Scrum Master:

Planning Sprint – Emmi

Sprint 1 – Farnaz

Sprint 2 – Emmi

Sprint 3 – Farnaz

Sprint 4 – Emmi

Sprint 5 – Farnaz

Sprint 6 – Emmi

Development Cycle

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Product Backlog

Planning sprint 18 maalis – 25 maalis (3 työtehtävää) | 0 | 3 | 0 | Päätä sprintti

APP-14	Project idea presentation	KÄYNNISSÄ	25.3	3	EK
APP-49	App layout design using Figma	KÄYNNISSÄ	25.3	-	FJ
APP-22	Project Plan Report	KÄYNNISSÄ	25.3	-	

+ Luo

Backlog (16 työtehtävää) | 35 | 0 | 0 | Luo sprintti

APP-1	Sprint-1-Planning	TEHTÄVÄLISTA	25.3	3	
APP-2	Sprint-2-Planning	TEHTÄVÄLISTA	1.4	3	
APP-3	Sprint-3-Planning	TEHTÄVÄLISTA	8.4	3	
APP-4	Sprint-4-Planning	TEHTÄVÄLISTA	15.4	3	
APP-5	Sprint-5-Planning	TEHTÄVÄLISTA	22.4	3	
APP-11	Sprint-6-Planning	TEHTÄVÄLISTA	29.4	3	
APP-6	Sprint-1-Retrospective	TEHTÄVÄLISTA	1.4	2	
APP-7	Sprint-2-Retrospective	TEHTÄVÄLISTA	8.4	2	
APP-8	Sprint-3-Retrospective	TEHTÄVÄLISTA	15.4	2	
APP-9	Sprint-4-Retrospective	TEHTÄVÄLISTA	22.4	2	
APP-10	Sprint-5-Retrospective	TEHTÄVÄLISTA	29.4	2	
APP-12	Sprint-6-Retrospective	TEHTÄVÄLISTA	6.5	2	
APP-13	Final presentation	TEHTÄVÄLISTA	6.5	5	
APP-23	As a user I want to see daily and weekly data of my dog's activity, so that I can review if my dog is getting enough exercise	TEHTÄVÄLISTA		-	
APP-30	As a user I want to store my dog's previous walks, so that I can track walks on long term	TEHTÄVÄLISTA		-	
APP-43	As a dog owner, I want the app to track my walk using GPS so that I can record my dog's distance, pace, and route accurately.	TEHTÄVÄLISTA		-	

+ Luo

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Product Backlog – User Story

As a new user, I want to create a dog profile so I can track my pet's activity.
As a user, I want to set daily walk goals so I can stay consistent.
As a user, I want a "Start Walk" button so I can begin tracking easily.
As a user, I want to see weather info, so I know if it's a good time to walk.
As a user, I want to see recent walks with distance and time so I stay motivated.
As a user, I want to see a live map of my walk route so that I can recognize the runs by area.
As a user, I want to filter history by week or day.
As a dog owner, I want the app to track my walk using GPS so that I can record my dog's distance, pace, and route accurately.
As a user, I want to compare runs to recommended activity, so that I know If my dog is getting enough exercise.
As a user, I want to view and set my dogs profile breed, weight and age.
As a user, I want to edit, add and delete my dog's info anytime.

Team Roles

Approach

- Work divided into Backend and Frontend areas
- Responsibilities are shared between both team members

Logic & Data (Backend)

- GPS tracking and route logic
- Room database
- API integration (weather)
- ViewModel and data handling
- Unit testing

UI & UX (Frontend)

- UI design (Jetpack Compose)
- Map, history, and statistics screens
- Navigation
- Visual design and accessibility
- UI testing

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Timeline

Sprint	Dates	Focus	Tasks
Sprint 1	25.-1.4	Map, route	MVVM, screens, Maps API, GPS
Sprint 2	1.-8.4	Room database, history	Room DB, save walks data, show history
Sprint 3	8.-15.4	Statistics	Calculate stats, display charts, test
Sprint 4	15.-22.4	UI improvements	Material Design, UI improvements
Sprint 5	22.-29.4	Testing, bug fixing	Optional features, Unit/UI tests, fix bugs
Sprint 6	29.4-6.5	Documentation, UAT (Optional)	Documentation, User testing

Risks and Mitigation

Risk 1: GPS may give inaccurate or delayed location updates and this can cause messy routes or incorrect distance calculations.

Especially:

- Indoors
- Near tall buildings
- In bad weather
- When battery optimization restricts location updates

Mitigation:

- Minimum distance filter
- Ignoring low-accuracy points
- Test outdoors in different environments
- Allow user to restart tracking if GPS is unstable

Risk 2: Google Maps Integration Problems: map may not load API key (misconfiguration)

Mitigation:
- Set up API key early in the project
- Test on both emulator and real device
- Keep API key in local.properties (not in GitHub)

Risk 3: Time Management Problems: some features may take longer than expected.

Mitigation:
- Follow the 6-week timeline
- Prioritize core feature first: GPS- Maps- Room
- Add advanced features only if time allows; like AI or notifications
- Keep UI simple at first then publish later

Success Metrics

- All core features (MVP) work as expected
- UI is accessible and follows Material Design
- All core features (MVP) work as expected
- UI is accessible and follows Material Design
- Clean Room database stores and retrieves data correctly
- Unit tests pass and application is tested with at least 50% coverage

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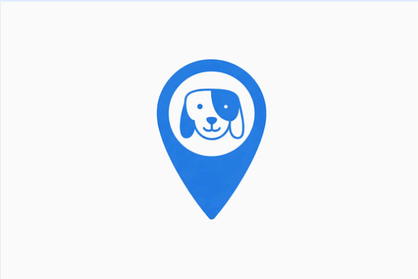
Questions

- Scrum role cycle weekly or time period?
- One product owner role or not?
- Logo/icon options?



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Thank You!

We appreciate feedback